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Apparatus and method for corneal refractive optimization using post-operative measurements

Abstract

The disclosure provides methods and apparatuses for determining a laser parameter set for corneal refractive surgery. The apparatus may include an autorefractor configured to obtain at least two ocular measurement parameters for an eye and to obtain a post-operative refraction of the eye. The apparatus may include a user interface configured to obtain a target refraction for the eye. The apparatus may include a memory and a processor communicatively coupled to the user interface, the autorefractor, and the memory. The processor may be configured to determine the laser parameter set based on an algorithm using the at least two ocular measurement parameters. The processor may be configured to correlate the at least two ocular measurement parameters, the laser parameter set, and the post-operative refraction as a training set.

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Background/Summary

CLAIM OF PRIORITY UNDER 35 U.S.C. § 119 (1) This application is a National Stage entry of the International Application No. PCT/US2020/035238, filed May 29, 2020, which claims priority to U.S. Provisional Application No. 62/855,364 titled “APPARATUS AND METHOD FOR CORNEAL REFRACTIVE OPTIMIZATION USING POST-OPERATIVE MEASUREMENTS,” filed May 31, 2019, which is assigned to the assignee hereof, and incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) Aspects of the present invention relate to systems, apparatuses, and methods for selecting a laser parameter set for corneal refractive surgery.

BACKGROUND

(2) Corneal refractive surgery may utilize a laser to ablate, remove, or shape a portion of the cornea to change a refraction of an eye. Example corneal refractive surgery techniques include laser-assisted in-situ keratomileusis (LASIK) and Small Incision Lenticule Extraction (SMILE). Multiple algorithms for determining laser parameters for corneal refractive surgery are currently available. For example, wavefront guided profile, a wavefront-optimized profile, or a topography-guided profile may be considered algorithms for determining laser parameters. Although the existing algorithms give similar results over a range of input parameters, they also diverge significantly at specified ranges of input parameters.

(3) Individual algorithms have been demonstrated to work best with certain input parameters. The input parameters may include ocular measurement parameters such as axial length, corneal power, a white-to-white distance, gender or sex, anterior chamber depth, pre-operative refraction, and/or lens thickness. For example, a particular algorithm may work better with “shorter” eyes and another particular algorithm may work better in “longer” eyes. Further, “adjustments” to these algorithms may be used to obtain better results. An “adjustment” may include any additional factor applied to a laser parameter calculation algorithm.

(4) The current state of the art includes selecting one algorithm to determine the laser parameters and possibly comparing the results to those obtained using another algorithm. A limited number of ophthalmologists understand the data and literature that support using one algorithm over another. While the use of a particular algorithm may be debatable, there are certain scenarios (e.g., a specific measured axial length or corneal power) in which one algorithm is generally accepted as better than others.

SUMMARY

(5) Aspects of the present disclosure may include apparatuses and methods for determining laser parameter sets for corneal refractive surgery. A first method may include determining one or more targets for corneal refractive surgery. The method may include obtaining at least two ocular measurement parameters. The method may include determining a laser parameter set based on an algorithm using the at least two ocular measurement parameters and the one or more targets for corneal refractive surgery. The method may include determining an estimated error of the algorithm using a deep learning machine trained on verified post-operative results. The method may include adjusting the targets for corneal refractive surgery based on the estimated error. The method may include redetermining the laser parameter set for corneal refractive surgery.

(6) A second method of laser parameter set selection may include obtaining at least two ocular measurement parameters for an eye by an autorefractor. The method may include determining one or more targets for corneal refractive surgery. The method may include determining a laser parameter set based on an algorithm using the at least two ocular measurement parameters. The method may include obtaining a post-operative refraction of the eye from the autorefractor. The

method may include correlating the at least two ocular measurement parameters, the laser parameter set, and the post-operative refraction as a training set.

(7) In another aspect, the disclosure provides a non-transitory computer-readable medium storing computer executable instructions. The instructions may be executed by a computer to obtain at least two ocular measurement parameters for an eye by an autorefractor or wavefront analyzer. The computer-readable medium may store instructions to determine one or more targets for corneal refractive surgery. The computer-readable medium may store instructions to determine a laser parameter set based on an algorithm using the at least two ocular measurement parameters. The computer-readable medium may store instructions to obtain a post-operative refraction of the eye from the autorefractor or the wavefront analyzer. The computer-readable medium may store instructions to correlate the at least two ocular measurement parameters, the laser parameter set, and the post-operative refraction as a training set.

(8) In another aspect, the disclosure provides an apparatus for determining a laser parameter set for corneal refractive surgery. The apparatus may include an autorefractor configured to obtain at least two ocular measurement parameters for an eye and to obtain a post-operative refraction of the eye. The apparatus may include a user interface configured to obtain a target refraction for the eye. The apparatus may include a memory and a processor communicatively coupled to the user interface, the autorefractor, and the memory. The processor may be configured to determine the laser parameter set based on an algorithm using the at least two ocular measurement parameters. The processor may be configured to correlate the at least two ocular measurement parameters, the laser parameter set, and the post-operative refraction as a training set.

(9) Additional advantages and novel features relating to aspects of the present invention will be set forth in part in the description that follows, and in part will become more apparent to those skilled in the art upon examination of the following or upon learning by practice thereof.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) In the drawings:

(2) FIG. 1 illustrates various features of an example computer system for use in conjunction with aspects of the present disclosure.

(3) FIG. 2 illustrates an example system diagram of various hardware components and other features for use in accordance with aspects of the present disclosure.

(4) FIG. 3 illustrates an example input user interface for use in accordance with aspects of the present disclosure.

(5) FIG. 4 is a diagram of an example computer system according to an aspect of the present disclosure.

(6) FIG. 5 is a flowchart of an example method for recommending an ablation profile according to an aspect of the disclosure.

(7) FIG. 6 is a diagram of an example apparatus according to an aspect of the present disclosure.

(8) FIG. 7 is a flowchart of a second example method for recommending an ablation profile according to an aspect of the disclosure.

(9) FIG. 8 is a conceptual diagram illustrating an example use context for the example apparatus of FIG. 6.

DETAILED DESCRIPTION

(10) Aspects of the present disclosure may include systems, apparatuses, and methods for determining a laser parameter set for corneal refractive surgery, which may include laser-assisted in-situ keratomileusis (LASIK), Small Incision Lenticule Extraction (SMILE), or other procedures that ablate, remove, or shape a portion of the cornea. A laser parameter set may refer to any

parameters used to configure a device for performing a corneal refractive surgery. For example, in a LASIK procedure using an excimer laser, the laser parameter set may be referred to as an ablation profile and may include one or more of: ablation diameter, ablation depth, ablation blend zone, corneal flap thickness, residual corneal bed, excimer fluence level, femtosecond laser flap size, or which femtosecond laser to use for flap. As another example, for a SMILE procedure, the laser parameter set may include one or more of: lenticule size/diameter, lenticule thickness, lenticule profile, corneal roof thickness, residual corneal bed, lenticule extraction tunnel size, femtosecond laser energy setting, or femtosecond laser spot size/spacing. In an aspect, post-treatments results may be utilized to train a machine-learning model (e.g., a neural network) to predict an error (e.g., a difference between a target and a post-treatment result) due to use of an existing algorithm for determining the laser parameter set. The parameters of the algorithm may be modified based on the predicted error to reduce the error. The post-treatment results may be used to further train the machine-learning and improve results.

(11) Aspects of the present disclosure may be implemented using hardware, software, or a combination thereof and may be implemented in one or more computer systems or other processing systems. In an aspect of the present disclosure, features are directed toward one or more computer systems capable of carrying out the functionality described herein. An example of such a computer system **300** is shown in FIG. 1.

(12) Computer system **300** includes one or more processors, such as processor **304**. The processor **304** is connected to a communication infrastructure **306** (e.g., a communications bus, cross-over bar, or network). Various software aspects are described in terms of this example computer system. After reading this description, it will become apparent to a person skilled in the relevant art(s) how to implement aspects of the disclosure using other computer systems and/or architectures.

(13) Computer system **300** can include a display interface **302** that forwards graphics, text, and other data from the communication infrastructure **306** (or from a frame buffer not shown) for display on a display unit **330**. For example, the display interface **302** may forward a graphical rendering of a super surface from the processor **304** to the display unit **330**. Computer system **300** also includes a main memory **308**, preferably random access memory (RAM), and may also include a secondary memory **310**. The secondary memory **310** may include, for example, a hard disk drive **312** and/or a removable storage drive **314**, representing a floppy disk drive, a magnetic tape drive, an optical disk drive, a universal serial bus (USB) flash drive, etc. The removable storage drive **314** reads from and/or writes to a removable storage unit **318** in a well-known manner. Removable storage unit **318** represents a floppy disk, magnetic tape, optical disk, USB flash drive, etc., which is read by and written to removable storage drive **314**. As will be appreciated, the removable storage unit **318** includes a computer usable storage medium having stored therein computer software and/or data.

(14) Alternative aspects of the present disclosure may include secondary memory **310** and may include other similar devices for allowing computer programs or other instructions to be loaded into computer system **300**. Such devices may include, for example, a removable storage unit **322** and an interface **320**. Examples of such may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an erasable programmable read only memory (EPROM), or programmable read only memory (PROM)) and associated socket, and other removable storage units **322** and interfaces **320**, which allow software and data to be transferred from the removable storage unit **322** to computer system **300**.

(15) Computer system **300** may also include a communications interface **324**. Communications interface **324** allows software and data to be transferred between computer system **300** and external devices. Examples of communications interface **324** may include a modem, a network interface (such as an Ethernet card), a communications port, a Personal Computer Memory Card International Association (PCMCIA) slot and card, etc. Software and data transferred via communications interface **324** are in the form of signals **328**, which may be electronic,

electromagnetic, optical or other signals capable of being received by communications interface **324**. These signals **328** are provided to communications interface **324** via a communications path (e.g., channel) **326**. This path **326** carries signals **328** and may be implemented using wire or cable, fiber optics, a telephone line, a cellular link, a radio frequency (RF) link and/or other communications channels. In this document, the terms “computer program medium” and “computer usable medium” are used to refer generally to media such as a removable storage drive **314** and a hard disk installed in hard disk drive **312**. These computer program products provide software to the computer system **300**. Aspects of the present disclosure are directed to such computer program products.

(16) In an aspect, the computer system **300** may include an ocular measurement device **350**. The ocular measurement device **350** may determine one or more ocular measurement parameters. An ocular measurement device may include any device for measuring an eye. For example, the ocular measurement device **350** may measure corneal power (keratometry), corneal topography, corneal tomography, corneal pachymetry, refraction, cyclopegia refraction, and wavefront refraction. Additionally, the ocular measurement device **350** may include an ocular measurement device such as a wavefront analyzer, which may perform corneal topography. The ocular measurement device **350** may further receive input of ocular measurement parameters (e.g., gender or sex). In some implementations, the ocular measurement device **350** may measure an axial length and a corneal power of an eye. In an aspect, the ocular measurement device **350** may further measure a white-to-white distance, anterior chamber depth, pre-operative refraction, and/or lens thickness. The axial length may be a distance from the surface of the cornea to the retina. The corneal power may be a dioptric power of the cornea. As another example, the ocular measurement device **350** may measure an anterior chamber depth of an eye. In an aspect, the ocular measurement device **350** may be an ultrasound device. In another aspect, the ocular measurement device **350** may be an optical biometer. Various optical biometers are available under the names LENSTAR® and IOL MASTER. In another aspect, the ocular measurement device **350** may include an intraoperative aberrometry device. The intraoperative aberrometry device may take measurements of refractive properties of the eye during surgery. For example, an intraoperative aberrometry device may provide information on sphere, cylinder, and axis of the eye. The ocular measurement device **350** may be communicatively coupled to the processor **304** via the communication infrastructure **306**, the communications interface **324**, and/or the communications path **326**.

(17) Computer programs (also referred to as computer control logic) are stored in main memory **308** and/or secondary memory **310**. Computer programs may also be received via communications interface **324**. Such computer programs, when executed, enable the computer system **300** to perform the features in accordance with aspects of the present disclosure, as discussed herein. In particular, the computer programs, when executed, enable the processor **304** to perform the features in accordance with aspects of the present disclosure. Accordingly, such computer programs represent controllers of the computer system **300**.

(18) In an aspect of the present disclosure where the disclosure is implemented using software, the software may be stored in a computer program product and loaded into computer system **300** using removable storage drive **314**, hard disk drive **312**, or communications interface **320**. The control logic (software), when executed by the processor **304**, causes the processor **304** to perform the functions described herein. In another aspect of the present disclosure, the system is implemented primarily in hardware using, for example, hardware components, such as application specific integrated circuits (ASICs). Implementation of the hardware state machine so as to perform the functions described herein will be apparent to persons skilled in the relevant art(s).

(19) In yet another aspect of the present disclosure, the disclosure may be implemented using a combination of both hardware and software.

(20) FIG. 2 shows a communication system **400** usable in accordance with aspects of the present disclosure. The communication system **400** includes one or more accessors **460** (also referred to

interchangeably herein as one or more “users”) and one or more terminals **442** and/or other input device or devices (e.g., an ocular measurement device **466**). In an aspect, the ocular measurement device **466** may be similar to the ocular measurement device **350** (FIG. 3). The ocular measurement device **466** may further be configured to communicate with the network **444**. In one aspect of the present disclosure, data for use is, for example, input and/or accessed after being received from an input device by accessors **460** via terminals **442**, such as personal computers (PCs), minicomputers, mainframe computers, microcomputers, telephonic devices, or wireless devices, personal digital assistants (“PDAs”) or a hand-held wireless devices (e.g., wireless telephones) coupled to a server **443**, such as a PC, minicomputer, mainframe computer, microcomputer, or other device having a processor and a repository for data and/or connection to a repository for data, via, for example, a network **444**, such as the Internet or an intranet, and/or a wireless network, and couplings **445**, **446**, **464**. The couplings **445**, **446**, **464** include, for example, wired, wireless, or fiberoptic links. In another aspect of the present disclosure, the method and system of the present disclosure may include one or more features that operate in a stand-alone environment, such as on a single terminal.

(21) In an aspect, the server **443** may be an example of the computer system **300** (FIG. 1). In an aspect, for example, the server **443** may be configured to perform the methods described herein. For example, the server **443** may obtain measurements such as an axial length and corneal power measurements from a terminal **442** and/or other input device. The server **443** may also determine one or more targets for corneal refractive surgery. For example, the measurements or the targets may be entered by an accessor **460**, or provided by an ocular measurement device **350** (FIG. 1). The server **443** may determine a laser parameter set based on an algorithm using the at least two ocular measurement parameters and the one or more targets for corneal refractive surgery. Further, the server **443** may determine an estimated error of the algorithm using a deep learning machine trained on verified post-operative results. The server **443** may adjust the targets for corneal refractive surgery based on the estimated error and redetermine the laser parameter set for corneal refractive surgery.

(22) FIG. 3 illustrates an example input user interface (UI) **500** for use in accordance with aspects of the present invention. The user interface **500** may be implemented by the server **443** and displayed on the terminal **442**, for example. The user interface **500** may allow a user to enter pre-operative measurements of one or more eyes as well as target parameters. The user interface **500** allows simultaneous calculation and plotting of both eyes. Input-data may include the corneal power (keratometry), corneal topography, corneal tomography, corneal pachymetry, refraction, cyclopegia refraction, or wavefront refraction. Slight variations of the pre-operative measurements may be used by different surgeons or in different countries. The desired postop refraction are then selected and the optimized calculations are then performed. The target refraction may be a goal specified by the surgeon. For example, a target refraction of 0.0 may be used.

(23) The user interface **500** may include an input field **510** for a right eye and an input field **520** for a left eye. Each input field **510**, **520** may include input fields for specific measurements or parameters. For example, the input field **510** may include corneal power (keratometry) field **511**, corneal topography field **512**, corneal tomography field **513**, corneal pachymetry field **514**, refraction field **515**, which may include one or both of cyclopegia refraction or wavefront refraction. The input field **510** may also include targets for corneal refractive surgery including a target refraction field **517**. The input fields **510**, **520** may also include a help icon (e.g., “?”) that provides a description of the measurement or parameter including allowed ranges. Some fields may use a drop-down menu to select a value.

(24) Additionally, the user interface **500** may include a surgeon field **530**, a patient field **531**, and a patient ID field **532**. The server **443** may generate records for the surgeon and patient based on the fields **530**, **531**, **532**. The user interface **500** may also include an import option **540** that may allow a user to upload a file (e.g., a spreadsheet) including measurements and parameters for one or more

patients. A dedicated toric calculator and a post-LASIK calculator may also be included.

(25) Turning now to FIG. 4, an example system **900** may recommend a laser parameter set for corneal refractive surgery based on an algorithm and a trained deep learning machine **912**. The system **900** may be implemented on a server **443**, for example. The system **900** may communicate with one or more terminals **442** via the user interfaces **500** discussed above. The system **900** may include an algorithm component **910** for determining a laser parameter set based on an algorithm using at least two ocular measurement parameters, a deep learning machine **912** such as neural network **920** for determining an estimated error of the algorithm, and the user interface **500** for obtaining at least two ocular measurement parameters and a target for corneal refractive surgery. The algorithm component **910** may further adjust the target for corneal refractive surgery based on the estimated error and redetermine the laser parameter set based on the algorithm and the adjusted target for corneal refractive surgery. The system **900** may further include one or more training sets **930**. The training sets **930** may include sets of post-operative data including two or more of the pre-operative measurements and parameters (e.g., corneal power (keratometry) field **511**, corneal topography field **512**, corneal tomography field **513**, corneal pachymetry field **514**, refraction field **515**, which may include one or both of cyclopegia refraction or wavefront refraction). The training sets **930** may include the selected laser parameter set, and one or more target parameters (e.g., post-operative refraction). The training sets **930** may be used to train one or more of the deep learning machine **912** for estimating an error of a laser parameter set determined by the algorithm component **910**, as explained in further detail below.

(26) The system **900** may include an administrative portal **940** for controlling access to the system **900**. For example, the administrative portal **940** may permit an administrative user to generate training sets **930** from verified results **960**. The verified results **960** may be uploaded in the form of a database or spreadsheet. The administrative user may select combinations of measurements and parameters to use for the training sets **930**. The administrative user may combine the uploaded verified results with any existing training sets **930**. The system **900** may generate a new neural network **920** based on a new or updated training set **930**. The system **900** may provide the administrative user with statistics regarding the neural network. For example, a neural network may be associated with input boundaries and correlation values. The administrative user may also configure access controls **950** to manage user accounts for different end users. The user accounts may be associated with saved patient data. Additionally, the user accounts may be associated with a customized neural network **920**. For example, a customized neural network **920** may trained with verified results **960** exclusively from a particular surgeon, practice group, or laser manufacturer. A customized neural network **920** may help control for unknown or immeasurable factors affecting the particular surgeon, practice group, or laser manufacturer.

(27) The algorithm component **910** may implement a laser parameter set determination algorithm. A laser parameter set determination algorithm may include any deterministic technique for generating a laser parameter set based on two or more ocular measurements. For example, the algorithm component **910** may include software executed by a processor to determine a laser parameter set according to an algorithm using input values from the user interface **500**. For example, the algorithm component **910** may implement one or more of: a wavefront guided profile, a wavefront-optimized profile, or a topography-guided profile. The algorithm component **910** may provide the determined laser parameter set to the neural network **920** along with all of the input measurements and parameters. In an aspect, the algorithm component **910** may be implemented as a machine learned algorithm.

(28) The learning machine **912** may use deep learning techniques to predict error of the algorithm component **910** based on the training set **930** including post-operative results. The learning machine **912** may be implemented by, for example, a neural network **920**, which may utilize a Python based tensor flow. The neural network **920** may have a number of hidden layers, each including a number of neurons. The parameters of the neural network **920** may be selected based

on results for a particular prediction. In an aspect, the learning machine **912** may include a computer processor (e.g., processor **304** that is programmed to execute instructions for developing the neural network **920** based on a network structure (e.g., number and type of layers). Once the learning machine **912** has trained the neural network **920** (or other learning machine), the processor configured with the trained learning machine **912** may determine the predicted error of the algorithm component **910** based the ocular measurement parameters. The neural network **920** may receive multiple numeric inputs to predict a single numeric output. In an implementation, the neural network may receive three numeric inputs (corneal power (keratometry), corneal topography, corneal tomography, corneal pachymetry, refraction, cyclopegia refraction, and wavefront refraction) and output an error value. The neural network **920** may be trained by one or more of the training set **930**. The training sets **930** may be considered labelled data because the training sets **930** may include the post-operative refraction, which may be used to determine the accuracy or error of the algorithm. Accordingly, when the neural network **920** receives the set of numeric inputs, the neural network **920** may predict an estimated error of the algorithm component **910**. The learning machine **912** may be implemented using different types of learning machines. For example, the learning machine **912** may use any combination of supervised and unsupervised learning techniques. The learning machine **912** may be structured as, for example, an artificial neural network, convolutional neural network, Bayesian network, or other deep learning model.

(29) The algorithm component **910** may then adjust the algorithm inputs according to the predicted error. In particular, the algorithm component **910** may adjust a target (e.g., target post-op refraction) based on the predicted error. For example, the new target refraction may be set to the difference between the user input target refraction and the neural net predicted error. As an example, for an eye with a set of ocular measurement parameters and a target refraction, the neural network **920** may predict an error of 0.25. That is, when the algorithm component **910** calculates a laser parameter set for an eye with the set of ocular measurement parameters and target refraction of -0.5 , then the eye will get re-calculated using the current algorithm component **910** but using the new target refraction value of $-0.5 - 0.25 = -0.75$ instead of the user input target refraction. In other words, the algorithm component **910** may adjust one component of the algorithm's input (e.g., target refraction) by subtracting the neural network predicted error from the input value.

(30) In an aspect, the learning machine **912** may identify one or more elements of a laser parameter set associated with an error. For example, the learning machine **912** may perform regression analysis on each element of the laser parameter set to determine whether any of the elements are more heavily correlated with an error. The learning machine **912** may suggest an adjustment to the algorithm itself to address an individual element of the laser parameter set.

(31) In an aspect, the algorithm component **910** may limit the adjustment to the algorithm by the neural network predicted error. For example, the neural network **920** may produce an extreme value in the case of an out-of-bounds case where the neural network **920** does not have good training data. The algorithm component **910** may limit the value of the predicted error. For example, the algorithm component **910** may limit the neural network predicted error never to exceed ± 0.5 diopters.

(32) In an aspect, the neural network **920** may be adjusted based on a new ocular measurement. For example, in an implementation, the neural network **920** was provided with both pre-operative and post-refractive measurements of patients who had previously had laser-assisted in-situ keratomileusis (LASIK). The post-refractive measurements can be viewed as an error in the algorithm due to the previous refractive surgery. Being trained based on the difference for the pre-operative measurements and post-refractive measurements, the neural network **920** may provide a correction to the result provided by the algorithm component **910**.

(33) FIG. 5 is a flowchart illustrating an example method **1000** of providing a laser parameter set for corneal refractive surgery. The method **1000** may be performed by the system **900**.

(34) In block **1010**, the method **1000** includes determining one or more targets for corneal

refractive surgery. In an aspect, for example, the UI **500** may determine the one or more targets for corneal refractive surgery. In an implementation, the UI **500** may obtain the one or more targets for corneal refractive surgery from an input by an operator. For example, the operator may input a target refraction (e.g., 0 diopters).

(35) In block **1020**, the method **1000** includes obtaining at least two ocular measurement parameters. In an aspect, for example, the UI **500** may obtain the at least two measurement parameters, for example, as input from the operator. In another implementation, the measurement parameters may be obtained from an ocular measurement device **466**. The ocular measurement parameters may include, for example, axial length, corneal power, corneal power index, and anterior chamber depth. In an aspect, the ocular measurement parameters may include intraoperative aberrometry measurements such as sphere, cylinder, and axis of the eye. In an aspect, the ocular measurement parameters may include one or more of corneal power (keratometry), corneal topography, corneal tomography, corneal pachymetry, refraction, cycloopia refraction, or wavefront refraction.

(36) In block **1030**, the method **1000** may include determining a laser parameter set based on an algorithm using the at least two ocular measurement parameters and the targets for corneal refractive surgery. For example, the algorithm component **910** may determine the laser parameter set based on the algorithm using the at least two ocular measurement parameters.

(37) In block **1040**, the method **1000** may include determining an estimated error of the algorithm using a deep learning machine trained on verified post-operative results. In an aspect, for example, the neural network **920** may determine the estimated error of the algorithm. The neural network **920** may have been trained on training sets **930** including verified post-operative results including post-operative refractions corresponding to laser parameter sets. The verified post-operative results may be obtained from a measurement device such as an autorefractor or a wavefront analyzer.

(38) In block **1050**, the method **1000** includes adjusting the targets for corneal refractive surgery based on the estimated error. In an aspect, for example, the algorithm component **910** may adjust the targets for corneal refractive surgery based on the estimated error. For instance, the algorithm component **910** may subtract the estimated error from a user input target based on the estimated error.

(39) In block **1060**, the method **1000** includes redetermining the laser parameter set for corneal refractive surgery using the at least two ocular measurement parameters and the adjusted targets. In an aspect, for example, the algorithm component **910** may redetermine the laser parameter set for corneal refractive surgery using the at least two ocular measurement parameters and the adjusted targets.

(40) Turning now to FIG. **6**, an example apparatus **1100** may combine various portions of the system **900** with medical diagnostic equipment to provide a single apparatus that implements all or part of the system **900**. For example, the apparatus **1100** may recommend a laser parameter set based on an algorithm and a trained deep learning machine **1112**. It should be appreciated that while in an example implementation the apparatus **1100** may physically include multiple components within a single case **1190**, the apparatus **1100** may also be implemented as interconnected components, which may or may not be physically co-located. For example, in an aspect, verified results from multiple apparatuses **1100** may provide post-operative measurements to a network located database in addition to or instead of using the post-operative measurements locally. Some components of the apparatus **1100** may be implemented as computer-executable instructions stored on a computer-readable medium such as memory **1106**. The instructions may be executed by a processor **1104**. In an aspect, the processor **1104** and the memory **1106** may reside within the case **1190**. Additionally, the apparatus **1100** may include a display **1108**, which may display a user interface **1102**. The display **1108** may be a touch sensitive screen that receives input from a user. Alternative or additional input/output devices may be included.

(41) In an aspect, the example apparatus **1100** may include a biometer **1170** and an autorefractor

1180 for obtaining measurements of an eye. The biometer **1170** may obtain physical characteristics of the eye such as, but not limited to, corneal power, axial length, anterior chamber depth, corneal power index, a white-to-white distance, and/or lens thickness. The autorefractor **1180** may obtain optical measurements of the eye such as, but not limited to, the refraction of the eye, sphere, cylinder, and axis. In an aspect, the autorefractor **1180** may perform other vision assessment functions including aberrometry, topography, keratometry, and pupillometry. For example, the autorefractor **1180** may include a wavefront analyzer or be referred to as a wavefront analyzer. Conventionally, biometers and autorefractors are separate devices that are used for different purposes. For example, a biometer may be used to obtain measurements for selecting an intraocular lens, whereas an autorefractor may be used to estimate a patient's prescription for eyeglasses or contact lenses. In an aspect, the biometer **1170** and the autorefractor **1180** may be located in separate sensor heads of the apparatus **1100**. The apparatus **1100** may include a single chinrest for positioning the patient with respect to one of the sensor heads. Each sensor head may be moved with respect to the chinrest to position the head for obtaining the respective measurements. It should be appreciated that various alternative physical arrangements of the biometer **1170** and autorefractor **1180** may be constructed. In an aspect, for corneal refractive surgery, the biometer **1170** may be optional and all ocular measurement parameters may be obtained from the autorefractor **1180**.

(42) In an aspect, the biometer **1170** and the autorefractor **1180** may store measurements in a patient data storage **1160**. The patient data storage **1160** may be a computer memory, preferably a non-volatile computer memory such as a hard disc drive, solid state drive, EEPROM, etc. The biometer **1170** and the autorefractor **1180** may access a file of a patient in the patient data storage **1160** and directly record measurements. Such automatic recording may reduce manual transcription errors. In an alternative implementation, the patient data storage **1160** may be stored externally such as, for example, on a doctor's patient management system or a network storage system, in which case the apparatus **1100** may electronically communicate with the external storage.

(43) The apparatus **1100** may include a user interface **1102**. The user interface **1102** may guide a user (e.g., a technician) through operating the biometer **1170** and autorefractor **1180** to obtain measurements from a patient. The user interface **1102** may also include user interfaces similar to the user interface **500** (FIG. 3). The user interface **1102** may automatically import measurements into the input field **510** from the patient data storage **1160**. The user interface **1102** may receive input from the target refraction field **517**.

(44) The algorithm component **1110** may be similar to the algorithm component **910**. The algorithm component **1110** may receive the ocular measurements directly from the biometer **1170**, the autorefractor **1180**, or from the patient data storage **1160**. The algorithm component **1110** may implement any of the laser parameter set determination algorithms described herein or known in the art.

(45) The algorithm component **1110** may provide the determined laser parameter set to the learning machine **1112** along with all of the input measurements and parameters. The learning machine **1112** may be similar to the learning machine **912** and include, for example, a neural network **1120**. The learning machine **1112** may use deep learning techniques to predict error of the algorithm component **1110** based on the training set **1130** including post-operative results. In an aspect, the training set **1130** may include post-operative results received from the autorefractor **1180**. In an aspect, the training set **1130** may include post-operative results from only the autorefractor **1180** such that the trained learning machine **1112** is specific for the apparatus **1100**. That is, by training the learning machine **1112** based on input measurements and post-operative results from a single apparatus, the apparatus **1100** may be calibrated to correct for previous errors. In another aspect, the training set **1130** may be combined with other verified results **1162** such as results from other apparatuses **1100**, which may be remotely located. The administration portal **1140** may receive and authenticate the verified results **1162**, for example, from a trusted web service. In an aspect, the

access control **1150** may be accessed by a user via the user interface **1102** to specify which training set **1130** to use.

(46) The apparatus **1100** may include a laser **1192** that is used in performing corneal refractive surgery. For example, the laser **1192** may include an excimer laser, a femtosecond laser, or a combination of one or more of each type of laser. The laser **1192** may receive the final laser parameter set from the learning machine **1112**. The laser **1192** may perform ablation according to the final laser parameter set for a LASIK procedure.

(47) In another aspect, some functionality of apparatus **1100** may be performed via network service **1142**. For example, the algorithm component **1110** may be periodically updated based on results of machine learning performed remotely. For example, the system **900** may provide the network service **1142**. The system **900** may periodically generate an updated algorithm, e.g., based on learning machine **912**, and provide the updated algorithm to the algorithm component **1110** via the network service **1142**. In that case, the learning machine **1112** and training set **1130** may be remotely located (e.g., as learning machine **912** and training set **930**). The admin portal **1140** may be used to receive the updated algorithm component **1110**.

(48) In another aspect, the apparatus **1100** may retain a local learning machine **1112**, which may be trained by the network service **1142** or system **900**. The apparatus **1100** may transmit correlated pre-operative and post-operative measurements to the network service **1142** via the admin portal **1140**. The system **900** may then train a learning machine **912** based on a training set **930** including the correlated pre-operative and post-operative measurements of apparatus **1100**. The system **900** may then provide the trained learning machine **1112** to the apparatus **1100** for installation. Accordingly, the apparatus **1100** may utilize the trained learning machine **1112** without performing training and without accessing verified results of other apparatuses **1100**, which may include confidential data or protected health information (PHI).

(49) FIG. 7 is a flowchart illustrating an example method **1200** of providing a recommended laser parameter set for corneal refractive surgery. The method **1200** may be performed by the apparatus **1100**. The method **1200** may include some similar blocks to the method **1000**. It should be appreciated that the methods **1000** and **1200** may be combined. For brevity, description of some duplicate blocks is omitted. Further, as described above, the system **900** may perform some optional blocks of the method **1200**.

(50) In block **1210**, the method **1200** includes obtaining at least two ocular measurement parameters for an eye by an autorefractor. In an aspect, for example, the autorefractor **1180** may obtain the at least two ocular measurement parameters and a lens selection parameter for an eye. In an implementation, the autorefractor **1190** may obtain the parameters for both eyes of a patient. The ocular measurement parameters may include, for example, corneal power (keratometry), corneal topography, corneal tomography, corneal pachymetry, refraction, cyclopegia refraction, or wavefront refraction.

(51) In block **1220**, the method **1200** includes determining one or more targets for corneal refractive surgery. In an implementation, the user interface **1102** may determine the one or more targets for corneal refractive surgery. For example, the targets for corneal refractive surgery may be a target refraction for the eye following the corneal refractive surgery. The one or more targets for corneal refractive surgery may be entered by a technician or an ophthalmologist.

(52) In block **1230**, the method **1200** includes determining a laser parameter set based on an algorithm using the at least two ocular measurement parameters and the targets for corneal refractive surgery. For example, the algorithm component **1110** may determine the laser parameter set based on an algorithm using the at least two ocular measurement parameters and the targets for corneal refractive surgery.

(53) In block **1240**, the method **1200** may include obtaining a post-operative refraction of the eye from the autorefractor. In an aspect, for example, the autorefractor **1180** may obtain the post-operative refraction of the eye. In an aspect, the autorefractor **1180** may be coupled to the biometer

1170 (if used to obtain ocular measurement parameters) such that the same apparatus is used to obtain the at least two ocular measurement parameters and the post-operative refraction.

Additionally, the measurements may be stored in a common patient data storage **1160**.

(54) In block **1250**, the method **1200** may include correlating the at least two ocular measurement parameters, the laser parameter set, and the post-operative refraction as a training set. Since the biometer **1170** and the autorefractor **1180** are communicatively coupled, or the autorefractor **1180** is used to obtain both pre and post-operative measurements, the training set can be correlated directly from the devices without need for human data entry, which may result in transcription errors. Further, consistency may be improved by generating multiple data sets using a known pair of biometer **1170** and autorefractor **1180**, or the same device for pre and post-operative measurements.

(55) In block **1260**, the method **1200** optionally includes training a deep learning machine using the post-operative refraction of the eye and the laser parameter set to determine an estimated error of the algorithm. In an aspect, the apparatus **1100** may train the learning machine **1112** using the post-operative refraction of the eye and the laser parameter set. The learning machine **1112** may be trained to estimate the error of the algorithm component **1110** for a particular set of input parameters including the at least two ocular measurement parameters and the laser parameter set. The learning machine **1112** may be trained by providing the training sets labeled with the post-operative refraction as the result. It should be understood that the learning machine **1112** may be trained on training data from previous procedures. The at least two ocular measurement parameters for a current procedure may not be included in the training data because the post-operative refraction is not available. Once the post-operative refraction becomes available, the complete training set may be used to further train or retrain the learning machine **1112**. In an aspect, the block **1260** may be performed by an external system such as the system **900**, which may communicate with the apparatus **1100** via a network service **1142**. The apparatus **1100** may receive the trained learning machine **1112** via the network service **1142**.

(56) In block **1270**, the method **1200** may optionally include determining an estimated error of the algorithm using a deep learning machine trained on verified post-operative results including post-operative refractions corresponding to the laser parameter set. In an aspect, for example, the learning machine **1112** may determine the estimated error of the algorithm. As discussed above, the learning machine **1112** may have been trained on training sets **1130** or **930** including verified post-operative results including post-operative refractions corresponding to the laser parameter set.

(57) In block **1280**, the method **1200** may optionally include adjusting the targets for corneal refractive surgery based on the estimated error. In an aspect, for example, the algorithm component **1110** may adjust the targets for corneal refractive surgery based on the estimated error. For instance, the algorithm component **1110** may subtract the estimated error from a user input lens selection parameter.

(58) In block **1290**, the method **1200** may optionally include redetermining the laser parameter set based on the algorithm and the adjusted targets for corneal refractive surgery. In an aspect, for example, the algorithm component **1110** may redetermine the laser parameter set based on the algorithm and the adjusted targets for corneal refractive surgery.

(59) FIG. **8** is a conceptual diagram illustrating an example use context for the example apparatus **1100**. The apparatus **1100** may be referred to as a self-calibrating autorefractor or laser. The apparatus **1100** may automate collection of objective data that can be used to calibrate the apparatus **1100**. For example, laser parameter set determinations of the apparatus **1100** may be improved using deep learning to analyze post-operative results obtained via the autorefractor **1180**. The apparatus **1100** may continually improve as additional data is collected.

(60) While aspects of the present disclosure have been described in connection with examples thereof, it will be understood by those skilled in the art that variations and modifications of the aspects of the present disclosure described above may be made without departing from the scope

hereof. Other aspects will be apparent to those skilled in the art from a consideration of the specification or from a practice in accordance with aspects of the disclosure disclosed herein.

Claims

1. A method for determining a laser parameter set for corneal refractive surgery, comprising: determining one or more targets for corneal refractive surgery; obtaining at least two ocular measurement parameters of an eye; determining a laser parameter set based on an algorithm using the at least two ocular measurement parameters and the one or more targets for corneal refractive surgery; determining an estimated error of the algorithm using a deep learning machine trained on verified post-operative results; adjusting the one or more targets for corneal refractive surgery based on the estimated error; redetermining the laser parameter set for corneal refractive surgery; and configuring a laser to emit a laser beam based on the redetermined laser parameter set.
2. The method of claim 1, wherein the laser parameter set includes one or more of ablation diameter, ablation depth, ablation blend zone, corneal flap thickness, residual corneal bed, excimer fluence level, femtosecond laser flap size, or type of femtosecond laser used.
3. The method of claim 1, wherein the at least two ocular measurement parameters include at least two of: corneal power (keratometry), corneal topography, corneal tomography, corneal pachymetry, refraction, cyclopegia refraction, or wavefront refraction.
4. The method of claim 1, wherein the verified post-operative results include one or more of: post-operative inflammation, post-op corneal power (keratometry), post-op corneal topography, post-op corneal tomography, post-op corneal pachymetry, or post-op refraction.
5. The method of claim 1, further comprising training the deep learning machine using a training set including at least the two ocular measurement parameters of the eye and the verified post-operative results for the eye to determine the estimated error of the algorithm.
6. The method of claim 5, wherein the training set includes the one or more targets for corneal refractive surgery.
7. The method of claim 1, wherein the one or more targets for corneal refractive surgery includes a target refraction.
8. The method of claim 1, wherein the algorithm for determining the laser parameter set includes a wavefront guided profile, a wavefront-optimized profile, or a topography-guided profile, or adjustments thereto.
9. The method of claim 1, wherein the laser parameter set is for a laser-assisted in-situ keratomileusis (LASIK) procedure or Small Incision Lenticule Extraction (SMILE) procedure.
10. A method of laser parameter set selection, comprising: obtaining at least two ocular measurement parameters for an eye by an autorefractor; determining one or more targets for corneal refractive surgery; determining a laser parameter set based on an algorithm using the at least two ocular measurement parameters; configuring a laser to emit a laser beam based on the laser parameter set; obtaining a post-operative refraction of the eye from the autorefractor; correlating the at least two ocular measurement parameters, the laser parameter set, and the post-operative refraction as a training set; and training a deep learning machine using the post-operative refraction of the eye and the laser parameter set to determine an estimated error of the algorithm.
11. The method of claim 10, further comprising: determining an estimated error of the algorithm using a deep learning machine trained on verified post-operative results including post-operative refractions corresponding to laser parameter sets; adjusting the one or more targets for corneal refractive surgery based on the estimated error; and redetermining a final laser parameter set based on the algorithm and the adjusted one or more targets for corneal refractive surgery.
12. The method of claim 11, wherein correlating the at least two ocular measurement parameters, the laser parameter set, and the post-operative refraction as the training set comprises including the final laser parameter set in the training set.

13. The method of claim 10, wherein the one or more targets for corneal refractive surgery is a target refraction.
14. The method of claim 10, wherein the at least two ocular measurement parameters are selected from the group consisting of: corneal power (keratometry), corneal topography, corneal tomography, corneal pachymetry, refraction, cyclopegia refraction, and wavefront refraction.
15. The method of claim 10, wherein the algorithm for determining the laser parameter set includes one or more of: a wavefront guided profile, a wavefront-optimized profile, or a topography-guided profile, or adjustments thereto.
16. A non-transitory computer-readable medium storing computer executable instructions, comprising instructions to cause a computer to: obtain at least two ocular measurement parameters for an eye by an autorefractor or a wavefront analyzer; determine one or more targets for corneal refractive surgery; determine a laser parameter set based on an algorithm using the at least two ocular measurement parameters; determine an estimated error of the algorithm using a deep learning machine trained on verified post-operative results; adjust the one or more targets for corneal refractive surgery based on the estimated error; redetermine the laser parameter set for corneal refractive surgery; configure a laser to emit a laser beam based on the redetermined laser parameter set; obtain a post-operative refraction of the eye from the autorefractor or the wavefront analyzer; and correlate the at least two ocular measurement parameters, the laser parameter set, and the post-operative refraction as a training set.
17. An apparatus for determining a laser parameter set for corneal refractive surgery, comprising: an autorefractor configured to obtain at least two ocular measurement parameters for an eye and to obtain a post-operative refraction of the eye; a user interface configured to obtain a target refraction for the eye; a memory; and a processor communicatively coupled to the user interface, the autorefractor, and the memory, and configured to: determine the laser parameter set based on an algorithm using the at least two ocular measurement parameters; correlate the at least two ocular measurement parameters, the laser parameter set, and the post-operative refraction as a training set; and train a deep learning machine using the post-operative refraction of the eye and the laser parameter set to determine an estimated error of the algorithm.
18. The apparatus of claim 17, wherein the processor is configured to: determine an estimated error of the algorithm using a deep learning machine trained on verified post-operative results including post-operative refractions corresponding to laser parameter sets; adjust the target refraction for the eye based on the estimated error; and redetermine a final laser parameter set based on the algorithm and the adjusted target refraction for the eye.
19. The apparatus of claim 18, further comprising a laser for performing the corneal refractive surgery based on the final laser parameter set.
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